AECM: Attention-Enhanced Cross-Modality for Aerial Image and Video Translation With Vision Transformers

Cong Guo, Cuihua Lin, Wenhao Lin, Kan Ren, *Member, IEEE*, and Qian Chen

Abstract*—Cross-modality translation between infrared and visible images plays a critical role in remote sensing and geospatial data analysis, enabling robust Earth observation under diverse environmental conditions. However, in complex aerial scenes captured by unmanned aerial vehicles (UAVs), existing generative adversarial network (GAN) approaches often struggle to preserve the structural integrity of the source image while generating high-quality target images. To address this limitation, we propose an attention-enhanced cross-modality (AECM) translation framework based on vision transformers (ViT), aiming to improve the quality of cross-modality image translation. The AECM framework introduces three innovations: first, an attention-enhanced ViT discriminator is designed to focus on semantically critical regions through learnable weights, thereby mitigating the long-tail distribution problem prevalent in aerial scenes. Second, a sliding self-attention-based cross-similarity loss is proposed to achieve multiscale feature alignment from global to local perspectives. Third, a PatchNCE contrastive loss with learnable weight optimization is incorporated to enhance feature consistency between input and output images. Comprehensive experiments are conducted on multiple aerial image benchmark datasets, and the generalization of the proposed method is further validated on infrared-visible video translation datasets. Experimental results demonstrate that AECM consistently outperforms existing methods in qualitative and quantitative evaluations, achieving state-of-the-art (SOTA) performance. Moreover, the generated images exhibit superior transferability in downstream tasks such as object detection and image fusion, confirming the effectiveness of AECM in advancing cross-modal remote sensing applications. Codes are available at https://github.com/Tracygc/AECM

***Index Terms*—Aerial visible-to-infrared image translation (AVIT), contrastive learning (CL), infrared-to-visible image or video translation (IVIVT), self-attention.**

I. Introduction

IMAGE translation between visible and infrared modalities is a fundamental task in remote sensing, surveillance, and autonomous navigation [1]. Visible images provide rich semantic and color information, whereas infrared images offer robustness to lighting and weather variations, making them particularly valuable in low-visibility environments [2]. Aerial visible images (AVIs) are highly accessible but suffer from significant limitations due to their susceptibility to environmental conditions such as lighting and weather [3]. In contrast, aerial infrared images (AIIs) provide reliable all-weather data by sensing thermal emissions, but their acquisition is often prohibitively expensive and complex [4]. Translating visible to infrared images thus emerges as a vital strategy [5], [6], [7], [8]. This approach enhances the robustness of geospatial analysis systems without incurring the high cost of physical infrared data collection [9], [10]. By effectively bridging this modality gap, aerial visible-to-infrared translation (AVIT) has the potential to significantly augment data resources for downstream tasks such as object detection [11], [12], [13], [14], tracking, and monitoring [15], person reidentification [16], [17], and multimodal fusion without modal alignment [18], [19].

Recent advances in image-to-image (I2I) translation have demonstrated considerable potential for cross-modality aerial image generation, and the emergence of generative adversarial networks (GANs) [20] has driven significant progress in the field. With further research, disentangled representation learning has emerged as a key direction. This approach enables the decomposition of image representations into independent content and attribute spaces, which enhances the diversity and controllability of the output [9]. Methods such as multimodal unsupervised image-to-image translation (MUNIT) [21], diverse image-to-image translation (DRIT) [22], and contrastive learning for unpaired image-to-image translation (CUT) [23] leverage feature decoupling and contrastive learning (CL) to preserve semantic structure and enable multimodal outputs. Building on this, multimodal unpaired I2I approaches leverage the separation of content and style to effectively facilitate cross-modal transformation. This design is suitable for AVIT, where disentangling shared semantic structures from domain-specific styles is essential for generating realistic results [1], [5].

However, prevailing unpaired methods that rely on cycle-consistency constraints reduce the dependency on paired data but exhibit certain limitations. In the generation of AIIs, the strict bijective (one-to-one) mapping imposed by cycle consistency tends to cause the translated images to retain unrealistic textures and edges from the source visible domain, which

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compromises output fidelity [6], [22]. Moreover, aerial infrared imagery shows significant intramodal variation due to temperature and atmospheric factors. Although diversity is crucial for downstream robustness, conventional noise injection strategies often suffer from mode collapse and limited stylistic variety. The AVIT task presents several domain-specific challenges. First, visible and infrared modalities follow distinct physical principles, resulting in nonlinear feature relationships. Second, semantic structures such as buildings and roads must be preserved. Third, thermal patterns must conform to thermophysical laws, which texture transfer alone cannot guarantee. Fourth, images captured by unmanned aerial vehicles (UAVs) exhibit varying viewpoints, scales, and complex backgrounds, which complicate translation [1], [24], [25]. These challenges highlight the need for an adaptive I2I translation method tailored to aerial cross-modality scenarios.

To address these challenges, particularly in aerial scenarios, we propose an innovative framework named attention-enhanced cross-modality (AECM) for unpaired I2I translation, focusing on achieving high-quality translation between visible and infrared images. The AECM framework integrates three core components: an attention-enhanced vision transformers (ViT) discriminator, a cross-similarity loss based on sliding self-attention, and a learnable weight-optimized PatchNCE (LWO-NCE) contrastive loss. To mitigate the neglect of small objects due to long-tail distributions and complex UAV backgrounds, we design an attention-enhanced ViT discriminator with a learnable weighting mechanism. This component adaptively focuses on semantically important regions, which helps preserve structural details and improve semantic consistency. To handle nonlinear feature relationships and ensure multiscale content consistency under varying viewpoints and scales, we introduce a cross-similarity loss based on sliding self-attention. This loss constrains structural consistency from global to local levels by leveraging self-attention maps, leading to more natural content preservation aligned with human visual perception. To enforce thermophysical plausibility and balance content preservation with style transfer, we develop a LWO-NCE contrastive loss. By adaptively weighting feature correspondences in semantically critical areas, this loss ensures plausible thermal distributions while allowing flexibility in background regions. These components collectively enhance the capacity of AECM for content retention and style transfer across semantic, structural, and thermal dimensions.

Overall, the contributions of AECM are as follows.

1) We design an attention-enhanced ViT discriminator with a learnable weighting mechanism that adapts to focus on semantically important regions. This mitigates the small-object omission issue caused by long-tail distributions, enhancing the semantic consistency and detail quality of generated images.
2) We introduce a cross-similarity loss based on sliding self-attention, which constrains multiscale content consistency from global to local scales through self-attention maps, thereby making the content more aligned with human visual perception.
3) We develop a LWO-NCE contrastive loss that balances feature correspondence between semantic key regions and background areas through an adaptive weighting mechanism, achieving reasonable style transfer while preserving content.
4) We perform extensive experiments on public datasets for visible-to-infrared (V2I) and infrared-to-visible translation. The results in multiple benchmarks demonstrate the superiority of our approach over existing state-of-the-art (SOTA) methods.

In this article, Section II offers related work for our research. Our method is introduced in Section III. The experiments are presented in Section IV. Section V discusses limitations and future work. Finally, Section VI concludes this article.

II. Related Work

This section reviews key developments in general I2I translation, V2I translation, infrared-to-visible translation, and video-to-video translation.

A. General I2I Translation

I2I translation aims to learn mappings between source and target domains in supervised (paired) or unsupervised (unpaired) settings. Early supervised methods like Pix2Pix [26] required paired datasets, which are often unavailable. To address this, unsupervised CycleGAN [27] introduced cycle-consistency loss for unpaired translation. This bidirectional approach inspired DiscoGAN [28], DualGAN [29], and StarGAN [30] for multidomain tasks. Subsequent models such as BicycleGAN [31], MUNIT [21], and SICGAN [32] integrated variational autoencoders and adversarial style transfer to enable multimodal outputs and improve generalization. To reduce computational complexity, unidirectional I2I translation methods have been proposed. These models employ a single generator–discriminator pair and avoid source-domain reconstruction. DistanceGAN [33] and GcGAN [34] introduced distance consistency and geometric consistency constraints to guide translation. CUT [23] leveraged patch-level CL for content preservation, while F/LSeSim [35] employed self-similarity for spatial consistency. Despite these developments, bidirectional models remain computationally heavy and prone to texture artifacts. Unidirectional models often struggle with significant domain shifts or semantic fidelity. Furthermore, current constraints remain sensitive to alignment errors and intradomain variability. These limitations require domain-aware frameworks for specialized tasks like cross-modality aerial image translation.

B. V2I Image Translation

V2I translation is critical for expanding infrared datasets in aerial and drone imaging. Early methods addressed the domain gap via feature disentanglement for paired RGB-infrared generation [9]. To overcome the limitations of paired datasets, multimodal frameworks like MUNIT [21] and DRIT [22] decomposed images into content and style spaces for various synthesis. Recent works such as diverse and realistic

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aerial visible-to-infrared image translation (DR-AVIT) [1] and multiscale feature decomposition for cross-modality visible-to-infrared drone image translation (MSFD) [6] utilize dual encoders and multiscale decomposition to enhance realism. Representation disentanglement remains central to AVIT by modeling independent variation factors. Methods leveraging CL and spatial correlation [23], [35], [36] improve structural consistency. Furthermore, cross-modality disentanglement frameworks separate spectral features from content to synthesize realistic infrared characteristics [5]. AVIT remains challenging due to the nonlinear physical relationship between visible and infrared modalities. Current methods struggle to maintain semantic structures and generate thermophysically accurate patterns. UAV imagery adds complexity through viewpoint variations, scale changes, and cluttered backgrounds. Convolutional neural network (CNN) architectures are limited by local receptive fields, limiting global scene modeling [5]. These constraints highlight the necessity for adaptive frameworks that integrate long-range modeling capabilities, such as transformer blocks, while ensuring precise feature extraction.

C. Infrared-to-Visible Image Translation

Infrared-to-visible image translation aims to generate interpretable color images from infrared inputs, supporting applications such as surveillance, navigation, and scene understanding. Early approaches relied on colorization and texture transfer techniques [37], [38], which mapped infrared signals to visible color spaces using reference-based transformations. Deep learning advancements led to GAN-based methods [34], [39] and domain adaptation methods [40], [41] that improved realism and semantic consistency. Recent models like CUT [23] and F/LSeSim [35] integrated contrastive and spatial losses to preserve structural details. Significant challenges persist due to the fundamental discrepancies between modalities. Infrared images inherently lack the detailed semantic and color information required to reconstruct realistic textures or precise object boundaries. GAN-based methods often exhibit structural distortions and fail to maintain thermophysical plausibility. Colorization remains constrained by reference dependency and manual tuning, while fusion-based methods [42], [43] struggle to generate natural daytime-like visuals. These limitations highlight the need for more robust frameworks that integrate semantic guidance, physical constraints, and temporal modeling for effective infrared-to-visible translation.

D. Video-to-Video Translation

Video-to-video translation aims to generate realistic and temporally consistent video sequences across different domains [44]. Lai et al. [45] utilized deep recurrent networks to minimize temporal losses and improve stability. Dai et al. [46] addressed optical flow estimation issues through relation-based multiscale temporal consistency. Tulyakov et al. [47] decomposed motion and content using recursive networks to maintain continuity between adjacent frames. Furthermore, Wang et al. [48] constrained generated frame distributions to match the source domain. Despite these developments, these methods primarily rely on paired video datasets. Unpaired V2V translation emerged to address data scarcity. Recycle-GAN [49] integrated temporal cues with generative adversarial networks (GANs) using cycle and recursive losses. Similarly, Mocycle-GAN [50] employed optical flow estimation to preserve temporal coherence and structural consistency. Infrared-to-visible GAN (I2V-GAN) [51] enhanced spatiotemporal consistency through perceptual cycle and similarity losses. Rome [52] strengthened correspondences via cross-domain region similarity matching across local and global perspectives. However, these algorithms require robust temporal constraints between frames to ensure sequence coherence in the video domain.

III. Method

This section describes the architecture of each module of our algorithm and the formulation of the loss functions.

A. Framework Overview

In the task of unpaired cross-modality translation, we consider two unpaired frames: a source domain $\mathbb{X} = \{x\}$ and a target domain $\mathbb{Y} = \{y\}$, where each sample $x, y \in \mathbb{R}^{C\times H\times W}$ represents an image or video frame from its respective domain. For video translation, each video is divided into an ordered sequence of frames, denoted as $x = \{x_1, x_2, \ldots, x_N\}$ and $y = \{y_1, y_2, \ldots, y_N\}$, where x_n and y_n indicate the n-th frame. The objective is to learn a generator $G : \mathbb{X} \to \mathbb{Y}$ that maps a source input x to a synthesized target $\tilde{y}$. For video tasks, this formulation extends to a sequence $\tilde{y} = \{\tilde{y}_1, \ldots, \tilde{y}_N\}$ where each frame is processed as $\tilde{y}_n = G(x_n)$. We employ two discriminators D_m and D_a to distinguish the synthesized samples from real target data y. This adversarial framework enhances both perceptual fidelity and temporal consistency. As illustrated in Fig. 1, our AECM training procedure involves two discriminators, D_m and D_a, and a generator G. In addition to the GAN loss, three loss functions are introduced: a shifted-attention cross-similarity loss, a LWO-NCE loss, and a cross-similarity-based temporal consistency loss. The discriminator D_m modifies its output to form a multilayer perceptron (MLP) network, while D_a adjusts the attention between different contents of the video frame using learnable weights. Both discriminators are designed to assess whether the input image is real. To facilitate the training of our network, the first two loss functions constrain the spatial consistency of the generated outputs, while the third loss enforces temporal consistency between adjacent generated video frames.

B. Feature Extraction

By introducing the natural language processing-based transformer [53] into computer vision, the vision transformer (ViT) [54] demonstrated the effectiveness of attention mechanisms in visual feature extraction. We employ ViT pretrained on large-scale datasets for feature extraction, obtaining token embeddings for subsequent loss function calculations. For a 2-D image, ViT divides the image into N patches and converts these patches into a sequence of vectors. This sequence

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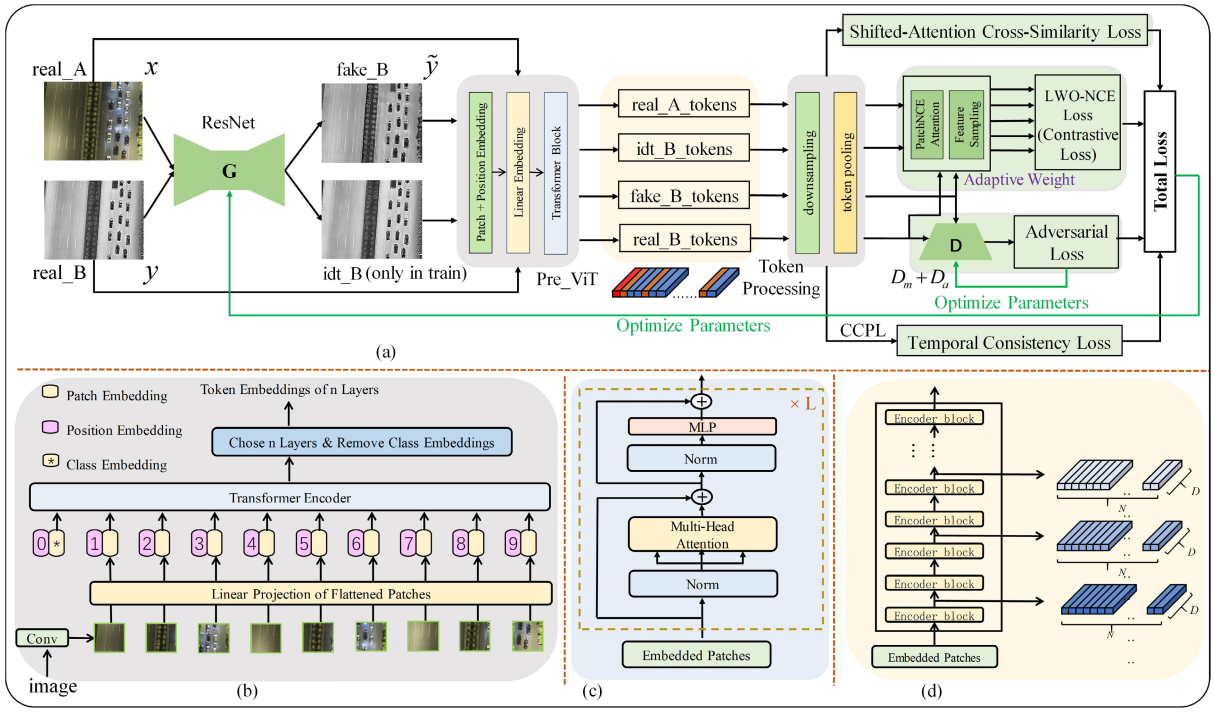


Fig. 1. Architecture of our AECM for cross-modality translation. "Pre_ViT" denotes a pretrained vision transformer. D_m and D_a represent the multiscale ViT discriminator and attention-enhanced ViT discriminator, respectively. "MLP" refers to multilayer perceptron, "CCPL" refers to contrastive coherence preserving loss, and "LWO-NCE Loss" refers to learnable-weight-optimized PatchNCE loss. (a) Architecture of our AECM. (b) Pre_ViT. (c) Vision transformer encoder. (d) Token embedding output.

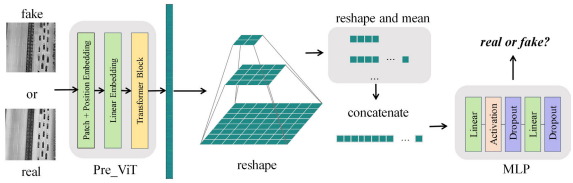


Fig. 2. Illustration of the multiscale ViT discriminator. The features from ViT are reshaped into different sizes for subsequent multiscale discrimination.

includes both class embeddings and positional embeddings. The sequence is input into a transformer encoder, which consists of L encoding blocks. Each block contains multihead attention sublayers and MLP sublayers. We adopt an encoder with $L = 12$ to process the image sequence. The standard ViT uses the token embeddings produced by the final encoder block, processes them through an MLP for classification, and outputs only the class embedding. In contrast, we extract token embeddings from multiple encoder blocks for downstream tasks. During the encoding process, we extract token embeddings from n layers, remove their class embeddings, and use the resulting features as final n-layer representations. Each layer feature consists of a sequence of length N and dimension D.

To construct a robust representation that preserves geometric details and captures semantic textures, we adopt a multilevel feature extraction strategy. Specifically, we extract token embeddings from the first, third, and fifth encoding blocks to serve as the output features, as shown in Fig. 1. The shallow layer first block) retains local patterns such as edges and textures, which are crucial for spatial precision. The middle layer (third block) transitions the representation from pixel-level details to part-level structures. The deeper layer (fifth block) captures global semantic context and is robust to local noise. By selecting nonadjacent layers, we implement a skip-layer sampling approach that maximizes information diversity and reduces interlayer redundancy compared to using consecutive blocks. Furthermore, empirical results from our ablation study in Section IV-E confirm that fusing these specific heterogeneous features yields the best balance between perceptual quality (IS) and structural fidelity (FID) for V2I image translation.

C. Two ViT Discriminators (Adversarial Loss)

As illustrated in Fig. 1, real and generated frames are fed into two discriminators to evaluate authenticity. During this process, each frame is processed by a pre-ViT discriminator.

1) Multiscale ViT Discriminator: ViT divides the image into patches of size $n \times n$. These patches are linearly projected and combined with positional embeddings to form tokens compatible with the transformer architecture. This approach addresses the issue of long sequence lengths in vision tasks using transformers. The global attention mechanism mitigates the limitation of convolutions that capture only local information. Our AECM integrates ViT with an MLP and proposes a multiscale ViT discriminator, D_m, as shown in Fig. 2.

Specifically, the translated frames and their corresponding target domain frames $(\tilde{y}, y)$ are separately fed into a pretrained ViT network. This yields token embeddings from a designated self-attention layer: T_y, $T_{\tilde{y}} \in \mathbb{R}^{N_r \times d}$, where $N_r = (H \times W)/(P^2)$

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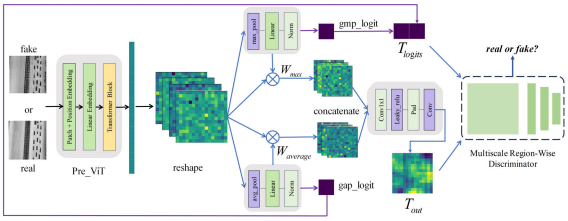


Fig. 3. Framework of the proposed attention-enhanced ViT discriminator.

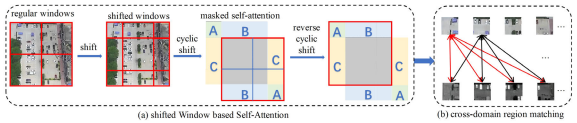


Fig. 4. Example of the shifted-window and the cross-domain region matching.

and $d = P^2 \times C$. Here, C is the number of channels and (P, P) is the resolution of each image patch (P = 16). These embeddings are then averaged in a nonoverlapping manner across multiple scales (specifically, 2, 4, 6, 8, and 12), producing new token embeddings such as $T_y^2, T_{\widetilde{y}}^2 \in \mathbb{R}^{(N_r)/(2)\times d}$, $T_y^4, T_{\widetilde{y}}^4 \in \mathbb{R}^{(N_r)/(4)\times d}$. Thus, we obtain $\mathbb{T}_y = T_y^2, T_y^4, \ldots, \mathbb{T}_{\widetilde{y}} = T_{\widetilde{y}}^2, T_{\widetilde{y}}^4, \ldots$. These embeddings are subsequently fed into the MLP discriminator for authenticity evaluation. The loss function is defined as in the following equation:

$$\mathcal{L}_{adv}^{D_m}(G, D_m) = \mathbb{E}_{y\sim P_{data(y)}}\left[logD_m(y)\right] + \mathbb{E}_{x\sim P_{data(x)}}\left[1 - logD_m(G(x))\right]. \quad (1)$$

2) Attention-Enhanced ViT Discriminator: In aerial drone datasets, backgrounds such as roads and vegetation occupy most pixels, while vehicles and pedestrians account for only a small fraction. During training, the network allocates more attention to backgrounds, reducing focus on foreground objects and causing a long-tail effect. To address this issue, we aim for the discriminator network to learn to distribute attention appropriately among different components of the frames. Therefore, we introduce learnable weights $\mathcal{W}$ and propose an attention-enhanced ViT discriminator, D_a.

Fig. 3 illustrates the framework for D_a. First, the token embeddings T_y, $T_{\widetilde{y}} \in \mathbb{R}^{N_r\times d}$ extracts from the pretrained ViT network, where $N_r = (H \times W)/(P^2)$ and $d = P^2 \times C$, are reshaped into $T_{y_r}, T_{\widetilde{y_r}} \in \mathbb{R}^{L\times P\times P}$. $L = N_r \times C$, C is the number of channels, and (P, P) is the resolution of each image patch (P = 16). Taking T_{y_r} as an example, global average pooling and global max pooling are applied to obtain $T_{y_avg}, T_{y_max} \in \mathbb{R}^{L\times 1\times 1}$. These are passed through their respective fully connected layers to produce two logits $gap_logit, gmp_logit \in \mathbb{R}^{1\times 1}$, which are concatenated into $T_{logits} \in \mathbb{R}^{1\times 2}$. Furthermore, the weights $\mathcal{W}_{y_avg}, \mathcal{W}_{y_max} \in \mathbb{R}^L$ are extracted from the fully connected layer parameters and applied to T_{y_r} to calculate the channel attention weighted feature maps ($T_{y_r}^{\mathcal{W}_{y_avg}}, T_{y_r}^{\mathcal{W}_{y_avg}} \in \mathbb{R}^{L\times P\times P}$). These feature maps are concatenated, followed by a 1×1 convolution and reflection padding, producing the output$T_{out} \in \mathbb{R}^{1\times(P-1)\times(P-1)}$ via a convolutional layer. Similarly, $T_{\widetilde{out}}, T_{\widetilde{logits}}$ are obtained. These pairs $(T_{out}, T_{\widetilde{out}}), (T_{logits}, T_{\widetilde{logits}})$ are used to calculate the MSE loss in (2). $D_a(y)$ calculates the combined MSE loss for T_{out} and T_{logits} derived from real frames, while $D_a(G(x))$ computes the loss for translated frames

$$\mathcal{L}_{adv}^{D_a}(G, D_a) = \mathbb{E}_{y\sim P_{data(y)}}\left[(D_a(y))^2\right] + \mathbb{E}_{x\sim P_{data(x)}}\left[1 - (D_a(G(x)))^2\right]. \quad (2)$$

D. Shifted-Attention Cross-Similarity Loss

We also propose the shifted-attention cross-similarity loss to investigate the impact of attention mechanisms. To capture both global and local information, we utilize a pretrained vision transformer (ViT) to apply global self-attention to features extracted from the generator's encoder layer, followed by two swin transformer blocks to enhance local self-attention. Finally, we compute the cross-similarity loss.

1) Shifted Window-Based Self-Attention: The self-attention mechanism of ViT is global. To further achieve local self-attention, the swin transformer [55] introduced a self-attention mechanism based on shifted windows. As shown in Fig. 4, the image is first divided into nonoverlapping regular windows, and self-attention is independently applied within each window. The windows are then shifted, and self-attention is computed across regions. Taking an 8×8 patch image as an example, it is divided into 2×2 windows of size 4×4, and the first self-attention calculation is performed. The windows are then shifted two patches to the right and downward, forming new window partitions. To facilitate computation, these windows are cyclically shifted and reassembled into regular patterns. The specially designed masks are then used to calculate self-attention. Subsequently, a reverse cyclic shift restores the windows to their original configuration. The swin transformer utilizes an even number of blocks, alternating between self-attention within regular windows and shifted windows. This approach enables the computation of local self-attention while approximating global self-attention, thereby increasing the amount of information captured.

2) Cross-Similarity Loss: To maintain spatial structural consistency between the generated images and the source domain, we calculate the cross-similarity between source and target domain images using a pretrained ViT. As shown in Fig. 4, the image is evenly divided into multiple patches, and the similarity mapping between each input patch and all output patches is calculated. Similarly, by interacting with input patches, a similarity map of the output patches is obtained. The calculation is (3). S_{s_i} represents the cross-similarity of the ith patch in the input image, and S_{t_i} denotes the cross-similarity of the i-th patch of the output image. $u_i, v_i \in \mathbb{R}^{1\times d}$ represent the ith patch of the input and the output, respectively. $u_*^T, v_*^T \in \mathbb{R}^{N_r\times d}$ denote all patches of the input and output, respectively. Therefore, considering all patches yields the sets of cross-similarities: $S_s = \{S_{s_1}, S_{s_2}, \ldots, S_{s_{N_r}}\}, S_t = \{S_{t_1}, S_{t_2}, \ldots, S_{t_{N_r}}\}$. By selecting results from several layers of ViT, we obtain $\mathbb{S}_s = \{S_s^{l^1}, S_s^{l^2}, \ldots, S_s^{l^n}\}, \mathbb{S}_t = \{S_t^{l^1}, S_t^{l^2}, \ldots, S_t^{l^n}\}$. The distance between the two sets is minimized to optimize training, as defined in (4), where $d(\cdot)$ denotes the cosine distance

$$S_{s_i} = u_i \times v_*^T \qquad S_{t_i} = v_i \times u_*^T \quad (3)$$

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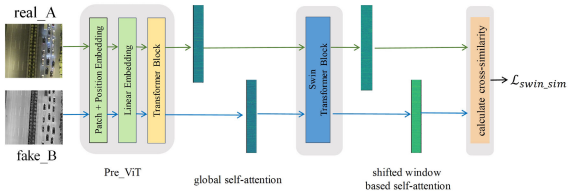


Fig. 5. Flowchart of calculating the shifted-attention cross-similarity loss.

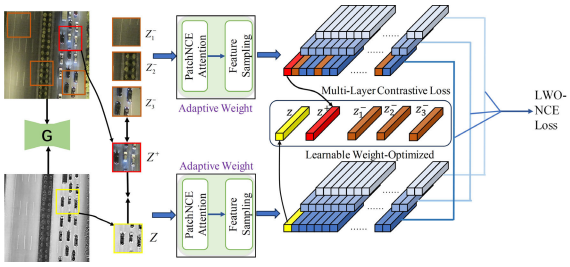


Fig. 6. Schematic of patchwise CL for LWO-NCE loss.

$$\mathcal{L}_{sim}(G) = \frac{1}{n} d(\mathbb{S}_s, \mathbb{S}_t) = \frac{1}{n} \sum_{j=1}^{n} d\left(S_s^{l^n}, S_t^{l^n}\right). \tag{4}$$

3) Shifted-Attention Cross-Similarity Loss: Cross-domain similarity is computed to enforce content and structural consistency between the source and target domains. This global alignment strategy enables the network to capture long-range dependencies across the entire spatial context. To further retrieve fine-grained local information, the image can be partitioned into nonoverlapping subregions for localized similarity calculation. Shifted-window self-attention is further computed using the outputs ($T_y, T_{\widetilde{y}} \in \mathbb{R}^{N_r \times d}$) of multilayer ViT. As illustrated in Fig. 5, two swin transformer blocks are employed following the ViT stage to extract local features, producing T_y^{swin} and $T_{\widetilde{y}}^{swin}$. This configuration complements the global self-attention of the pretrained ViT with the local window-based attention of the swin transformer. Finally, cross-similarity is calculated using T_y^{swin}, $T_{\widetilde{y}}^{swin}$, denoted as $\mathcal{L}_{swin_sim}$ in (5), where $T_y \to T_y^{swin}, T_{\widetilde{y}} \to T_{\widetilde{y}}^{swin}$

$$\mathcal{L}_{swin_sim}(G) = \frac{1}{n} d\left(\mathbb{S}_s^{swin}, \mathbb{S}_t^{swin}\right). \tag{5}$$

E. Learnable-Weight-Optimized PatchNCE Loss

CL facilitates the effective extraction of meaningful representations from unlabeled data, thereby supporting model training on unpaired datasets and enabling robust feature learning from frames. Building on this capability, we introduce a weight-optimized PatchNCE loss based on CL. Specifically, we propose a learnable weight optimization module that enhances salient features extracted from multiple encoder layers of the generator. This module emphasizes informative regions while suppressing less relevant ones. The contrastive loss computed using these refined features is denoted as the learnable-weight-optimized PatchNCE (LWO-NCE) loss.

1) Patchwise Contrastive Learning (PatchNCE): CL focuses on comparing the similarity or dissimilarity between samples. It requires that similar samples have minimal distance, whereas dissimilar samples have maximal distance. As shown in Fig. 6, it is essential not only to maintain spatial consistency between the entire input real frame and the output generated frame but also between their respective subdivided patches. In other words, patchwise CL is required. In CL, the primary components are the query and its corresponding positive, which are compared against other samples in the dataset, referred to as negatives. Let the vectors $z, z^+ \in \mathbb{R}^K$ denote the query and positive sample, while $z^- \in \mathbb{R}^{N \times K}$ denotes N negatives, and $z_n^- \in \mathbb{R}^K$ denotes to the n-th negative. This setting constitutes an (N+ 1)-way classification problem, where cross-entropy loss quantifies the probability of misclassifying a positive sample as negative. Equation (6) defines the PatchNCE loss, denoted as ℓ, where distances between the query and other examples are scaled by a temperature parameter $\tau = 0.07$

$$\ell\left(z, z^+, z^-\right) = -\log\left[\frac{\exp\left(z \cdot z^+/\tau\right)}{\exp\left(z \cdot z^+/\tau\right) + \sum_{n=1}^{N} \exp\left(z \cdot z_n^-/\tau\right)}\right]. \tag{6}$$

2) Weight-Optimized: When x is input to the generator, feature maps $\mathcal{Z} = \{z_1, z_2, \ldots, z_L\}$ are extracted from each of the L encoding layers, ranging from shallow to deep. These features are then weighted to obtain $\mathcal{Z}_W = \{z_{W1}, z_{W2}, \ldots, z_{WL}\}$. The generator is designed to learn how to effectively allocate attention to different parts of the frames during training. Our proposed weight optimization network $\mathcal{W}$ comprises six linear layers with spectral normalization, followed by activation layers. These linear layers are divided into two groups, each handling features of three different dimensions for real frames and fake frames, mapping the feature dimensions to 1. Specifically, we take the corresponding network layer weight parameters $\mathcal{W} = \{w_1, w_2, w_3\}$ from W and perform weighted operations on features of different sizes. As training progresses, these weight parameters $\mathcal{W}$ are updated and optimized iteratively. We use $\mathcal{S} = \mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_L$ to represent the set of spatial positions in each layer, where each position corresponds to a patch. For the lth layer, $Z_l^s \in \mathbb{R}^{C_l}$ denotes features of the corresponding patch, and $Z_l^{\mathcal{S}-s} \in \mathbb{R}^{(s_l-1)\times C_l}$ denotes features of the remaining patches. Similarly, when $\widetilde{y}$ is used as input, we obtain the weighted features $\widetilde{\mathcal{Z}}_W = \{\widetilde{z}_{W1}, \widetilde{z}_{W2}, \ldots, \widetilde{z}_{WL}\}$.

3) LWO-NCE Loss: After obtaining the weight-optimized features $\mathcal{Z}_W = \{z_{W1}, z_{W2}, \ldots, z_{WL}\}$ and $\widetilde{\mathcal{Z}}_W = \{\widetilde{z}_{W1}, \widetilde{z}_{W2}, \ldots, \widetilde{z}_{WL}\}$, patchwise CL is performed. For the l-th layer, when $\widetilde{z}_l^s$ is the feature of a specific patch in $\widetilde{y}$ (serving as the query), its positive sample z_l^s is the feature of the corresponding patch in x, while its multiple negative examples $z_l^{\mathcal{S}-s}$ are the features of the remaining patches in x. The objective is to maximize mutual information between $\widetilde{y}$ and x, ensuring structural consistency between them. For example, the shape of the tree in the generated image should match that in the target image. Based on these principles, we propose the LWO-NCE loss, denoted as $\mathcal{L}_{LWO_NCE}$, as shown in Fig. 6. The computation is provided in (7), where G represents the generator, W denotes the weight optimization, and the term $\widetilde{z}_l^s$, z_l^s, and $z_l^{\mathcal{S}-s}$ correspond to the query, its

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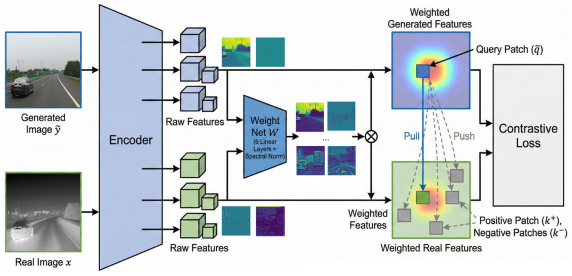


Fig. 7. Detailed structure diagram of the LWO-NCE. The loss function processes features through two streams: the generated stream (blue) and the real stream (green). The heatmap visualizes the learned weights W, which are applied to the features to emphasize semantically important regions. In the CL phase, solid arrows represent maximizing similarity between the query and the positive key, while dashed arrows represent minimizing similarity against negative keys within the weighted real features.

positive sample, and its negative samples, respectively

$$\mathcal{L}_{LWO_NCE}(G, W) = \mathbb{E}_{x\sim P_{data}(x)} \sum_{l=1}^{L} \sum_{S=S_1}^{S_1} \ell\left(\tilde{z}_l^s, z_l^s, z_l^{S-s}\right). \quad (7)$$

To explicitly distinguish semantic importance from feature intensity, the proposed weight optimization network W adaptively processes multiscale feature maps $\mathcal{Z}$. Rather than maximizing raw activation strength, the optimization of W is guided by the discriminative characteristics of the contrastive objective. In a contrastive setting, homogeneous or repetitive regions, such as flat backgrounds, even with high feature intensity, increase ambiguity among negative samples and hinder loss minimization. In contrast, semantically distinct regions, such as object boundaries and structural components, provide unique correspondences that facilitate alignment between the query and its positive sample. The weight-optimized network $\mathcal{W}$ assigns higher weights to semantically informative patches for accurate positive matching, while suppressing nondiscriminative background noise to reduce false negatives. The detailed architecture is shown in Fig. 7 using infrared images translated into the visible domain. The encoder extracts feature maps from both the real image x and the generated image $\tilde{y}$. These feature maps are processed by the weight-optimized network $\mathcal{W}$ to generate a spatial heatmap (attention map), which is multiplied element-wise with the original features to obtain refined weighted features. During contrastive loss computation, a query patch from the weighted generated features is pulled toward its corresponding positive patch in the real domain and pushed away from multiple negative patches, which enforces structural and semantic consistency.

F. Temporal Consistency Based on Cross-Similarity

It is crucial to maintain continuity between video frames due to the sequential nature of video data in cross-modality video translation. This approach resonates with the methodology proposed by contrastive coherence preserving loss (CCPL) [56] for versatile style transfer. In CCPL, random sampling of generated features is performed at various spatial locations across different feature layers, with a corresponding set of vectors sampled from the content features at the same positions. For each sampled vector, the objective is to preserve relational coherence with neighboring vectors. Rather than simply enforcing directional similarity of corresponding difference vectors, CCPL employs a CL strategy to maximize mutual information between matched difference vectors, while simultaneously pushing apart difference vectors from different locations within the latent space. This contrastive formulation mitigates the direct conflict between style adaptation and content preservation, enabling a more balanced integration of stylization and structural integrity in both images and videos. Our AECM integrates cross-similarity losses with contrastive coherence strategies. This approach ensures robust temporal and structural consistency, enhancing the fidelity and realism of cross-modality video translation and style transfer. Therefore, we propose a cross-similarity loss for temporal consistency, denoted as $\mathcal{L}_{temporal}$. Specifically, the cross-similarity is computed between the tth translated frame and the $(t+1)$th real frame, as well as between the (t+ 1)th translated frame and the tth real frame. Subsequently, the cosine distance between these cross-similarities is calculated, followed by the derivation of the L1 loss. The calculation is shown in (8), where N denotes the number of frames and n denotes the number of layers

$$\mathcal{L}_{temporal}(G) = \frac{1}{n} \sum_{t=1}^{N-1} d\left(\mathbb{S}_s(t, t+1), \mathbb{S}_t(t, t+1)\right). \quad (8)$$

G. Overall Optimization

Our proposed AECM framework integrates the generator, discriminators, and loss functions to address the challenges of unpaired cross-modality translation identified in Section I. To generate realistic textures while maintaining global semantic coherence in complex aerial scenes, the multiscale ViT discriminator (D_m) and the attention-enhanced ViT discriminator (D_a) work collaboratively. D_m enforces perceptual fidelity across multiple resolutions, and D_a uses learnable weights to focus adversarial feedback on semantically critical regions, which alleviates the long-tail distribution problem. In addition to adversarial objectives, the preservation of structural and semantic content is ensured by two synergistic constraints on the shared ViT feature space. The shifted-window cross-similarity loss ($\mathcal{L}_{swin_sim}$) establishes global-to-local alignment and maintains macro-structural consistency. The learnable-weight-optimized PatchNCE loss ($\mathcal{L}_{LWO_NCE}$) serves as an adaptive local constraint, emphasizing informative patches to preserve fine-grained details. The discriminators promote realistic modality-specific style synthesis, and the contrastive and similarity losses firmly anchor content to the source domain. This mechanism resolves the tradeoff between style transfer and content preservation. The overall objective of AECM is defined in (9) and (10), where $\lambda_1, \lambda_2, \lambda_3, \lambda_4, and \lambda_5$ denote the corresponding weight parameters

$$\begin{aligned}\mathcal{L}_D = &-\lambda_1 \mathbb{E}_{y\sim P_{data}(y)}\left[\log D_m(y)\right] \\ &- \lambda_1 \mathbb{E}_{x\sim P_{data}(x)}\left[1 - \log D_m(G(x))\right] \\ &- \lambda_2 \mathbb{E}_{y\sim P_{data}(y)}\left[(D_a(y))^2\right] \\ &- \lambda_2 \mathbb{E}_{x\sim P_{data}(x)}\left[1 - (D_a(G(x)))^2\right] \end{aligned} \quad (9)$$

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$$\begin{aligned}\mathcal{L}_G = &\lambda_1 \mathbb{E}_{x\sim P_{data}(x)}\left[1-\log D_m\left(G\left(x\right)\right)\right]\\ &+\lambda_2 \mathbb{E}_{x\sim P_{data}(x)}\left[1-\left(D_a\left(G\left(x\right)\right)\right)^2\right]\\ &+\lambda_3 \mathcal{L}_{swin_sim}\left(G\right)\\ &+\lambda_4 \mathcal{L}_{LWO_NCE}\left(G,W\right)\\ &+\lambda_5 \mathcal{L}_{temporal}\left(G\right).\end{aligned} \tag{10}$$

IV. Experiments Results

To validate the effectiveness of the proposed AECM, comparative experiments are conducted against SOTA algorithms using three infrared-to-visible (I2V) and two V2I datasets. Generated images are also evaluated on downstream tasks, specifically object detection and image fusion, to demonstrate practical utility.

A. Datasets

For V2I drone image translation, experiments utilize the Day-DroneVehicle and Night-DroneVehicle datasets introduced in DR-AVIT [1]. These datasets are derived from the DroneVehicle benchmark [12] by filtering invalid samples and categorizing data into daytime and nighttime conditions. Compared to earlier task-specific datasets such as the aerial visible-to-infrared image dataset (AVIID) [2], the DroneVehicle variants offer a larger volume of accurately paired images captured in diverse aerial scenarios. For evaluation, AVIID and both DroneVehicle subsets are randomly partitioned into training and testing sets using a 3:1 ratio. To ensure bidirectional coverage, AECM is further evaluated on infrared-to-visible benchmarks, including the infrared-to-visible image (IRVI) dataset [51] and the InfraredCity-Lite (ICL) dataset [52].

1) AVIID contains 1280 image pairs at 512×512 resolution comprising roads and parking lots [2]. We employ the AVIID3-paired-8 and AVIID3-unpaired-8 subsets for evaluation.

2) Day-DroneVehicle includes 7660 aligned V2I image pairs at 640×512 resolution captured by drones at varying altitudes and viewing angles during daytime [1].

3) Night-DroneVehicle consists of 13 140 pairs at 640×512 resolution under challenging low-light conditions [1].

4) IRVI provides video-based continuous sequences of traffic scenes at 256×256 resolution [51]. Evaluation utilizes 6045 frames from the Traffic subset and the rooftop_01 subset.

5) ICL contains urban and highway scenes across 603 142 frames [52]. We utilize the clear-day subset from the city and highway categories, focusing on the 256×256 resolution daytime visible and nighttime infrared frames.

B. Experiment Setup

The quality of generated images is evaluated through visual perception and task-based performance. Visual evaluation assesses the fidelity of synthesized infrared drone images relative to real data, while task-based evaluation measures performance in downstream applications.

1) Evaluation Metrics: Performance is quantified using five objective metrics: Fréchet inception distance (FID) [57], Kernel inception distance (KID) [58], learned perceptual image patch similarity (LPIPS) [59], structural similarity index (SSIM) [60], and Inception Score (IS) [20].

FID measures the realism of generated images by calculating the distance between deep feature distributions as shown in (11), where m and C represent the mean and covariance of inception features. $Tr(.)$ is the trace operation. It is based on the inception network [61], which has high stability, generalization, and robustness on small datasets. KID utilizes the maximum mean discrepancy (MMD) in embedding space with a cubic kernel to align with human perception. Unlike FID, KID does not assume normality and provides an unbiased estimate. MMD is formally defined as in (12), where P denotes the distribution of generated images and Q represents the distribution of real images. After derivations, the squared MMD provides an unbiased estimator as defined in (13), where k denotes the kernel function applied to m and n samples. Lower FID and KID values indicate superior content and structural similarity to real images

$$\begin{aligned}&\mathrm{FID}\left(Y,\hat{Y}\right)\\ &=\left\|m_Y-m_{\hat{Y}}\right\|_2^2+\mathrm{Tr}\left(C_Y+C_{\hat{Y}}-2\left(C_YC_{\hat{Y}}\right)^{\frac{1}{2}}\right)\end{aligned} \tag{11}$$

$$\begin{aligned}&MMD\left(\mathbb{P},\mathbb{Q};\mathcal{H}\right)\\ &=\sup_{f\in\mathcal{H},\|f\|_{\mathcal{H}}\le1}\mathbb{E}_{\mathbb{P}}f\left(X\right)-\mathbb{E}_{\mathbb{Q}}f\left(Y\right)\end{aligned} \tag{12}$$

$$\begin{aligned}&\mathrm{MMD}_u^2\left(X,Y\right)\\ &=\frac{1}{m\left(m-1\right)}\sum_{i\neq j}^{m}k\left(x_i,x_j\right)\\ &\quad+\frac{1}{n\left(n-1\right)}\sum_{i\neq j}^{n}k\left(y_i,y_j\right)-\frac{2}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}k\left(x_i,y_j\right).\end{aligned} \tag{13}$$

LPIPS evaluates perceptual similarity by extracting multilayer features through a pretrained CNN. The LPIPS distance $\mathcal{D}$ represents a weighted L_2 distance between normalized feature maps as formulated in (14). SSIM quantifies structural fidelity by comparing luminance, contrast, and structure within local windows according to (15). Lower LPIPS and higher SSIM values signify higher perceptual and structural integrity. IS assesses quality and diversity within the generated set using a pretrained Inception v3 network as per (16). In V2I translation, the generated IS is compared against the ground-truth (GT) IS. This relative assessment accounts for the domain gap between infrared data and the ImageNet dataset, as infrared images lack the saliency features of the pretraining data. Consequently, optimal performance is achieved when the generated IS closely approximates the GT IS

$$\mathcal{D}\left(\hat{x},x\right)=\sum_{l=1}^{L}\frac{1}{H_lW_l}\sum_{i,j}w_l\cdot\left\|\phi_l\left(\hat{x}\right)_{i,j}-\phi_l\left(x\right)_{i,j}\right\|_2^2 \tag{14}$$

$$\begin{aligned}\mathrm{SSIM}\left(x,y\right)=&\frac{\left(2\mu_x\mu_y+C_1\right)}{\left(\mu_x^2+\mu_y^2+C_1\right)}\cdot\frac{\left(2\sigma_x\sigma_y+C_2\right)}{\left(\sigma_x^2+\sigma_y^2+C_2\right)}\\ &\cdot\frac{\left(\sigma_{xy}+C_3\right)}{\left(\sigma_x\sigma_y+C_3\right)}\end{aligned} \tag{15}$$

$$IS\left(x\right)=\exp\left(\mathbb{E}_x\left[D_{KL}\left(p\left(y|x\right)\parallel p\left(y\right)\right)\right]\right). \tag{16}$$

2) Task Feedback: Generated images are evaluated through object detection feedback to assess their practical utility. For V2I drone translation, a YOLOv8 [62] model is trained on

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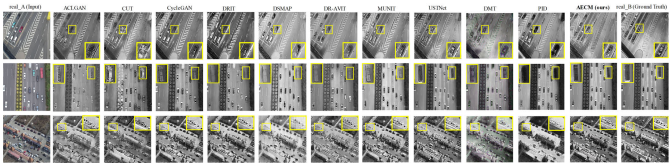


Fig. 8. Results comparison with different methods on Day-DroneVehicle.

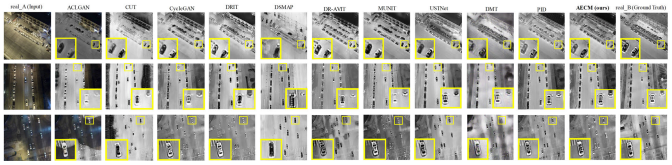


Fig. 9. Results comparison with different methods on Night-DroneVehicle.

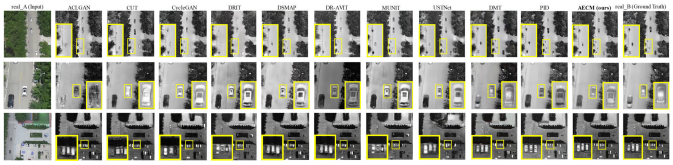


Fig. 10. Results comparison with different methods on AVIID3unpaired8.

authentic infrared images and tested on synthesized Day-DroneVehicle and Night-DroneVehicle datasets. Detection performance is quantified using mean average precision at a 0.5 intersection-over-union threshold (mAP@50) for the "car" category as defined in (17). Higher mAP values indicate superior recognition accuracy and synthesis quality. For infrared-to-visible translation, qualitative evaluation is conducted using a pretrained YOLOv5 [63] architecture

$$mAP = \left(\sum_{i=1}^{n} AP_i\right) \Big/ N. \tag{17}$$

3) Implementation Details: For aerial visible-infrared translation (AVIT), input images are cropped to 256×256 patches with horizontal flipping for data augmentation. Model optimization utilizes the Adam optimizer [64] with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. The learning rate is constant for 50 epochs before linearly decaying to zero over the final 50 epochs. FID and KID scores are computed via the Torch fidelity package [65] using subset sizes of 150 for AVIID and 1000 for DroneVehicle datasets. For infrared-to-visible image or video translation (IVIVT), the generator utilizes a ResNet_9blocks architecture across all comparative methods. Training is conducted for 100 epochs with a batch size of 1 and an initial learning rate of 1×10^{-5} followed by linear decay. Objective function weights for AECM are configured as $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = 5$, $\lambda_4 = 0.1$, $\lambda_5 = 0.1$.

C. Comparison Experiments for AVIT

To evaluate the performance of our AECM for AVIT, we conduct comparative experiments with SOTA methods.

1) Qualitative Results: Figs. 8–11 illustrate translation outcomes across AVIT benchmarks. Zoomed regions facilitate the inspection of small target reconstruction and structural edge

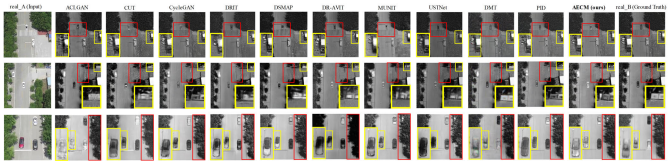


Fig. 11. Results comparison with different methods on AVIID3paired8.

fidelity. AECM outputs closely resemble authentic infrared data by preserving the geometric structure of targets. While methods such as CycleGAN and MUNIT often blur vehicle contours or introduce redundant artifacts, AECM maintains clear boundaries for small objects and ensures the integrity of large-scale structures like roads and buildings. Unlike CUT, our approach avoids geometric distortions. Although ACLGAN and DR-AVIT attempt content-style decoupling, they frequently retain source features or fail to match infrared tonal characteristics. On the AVIID-unpaired dataset, AECM translates high-quality visible inputs into infrared results that circumvent sensor-inherent motion artifacts, ensuring superior structural consistency. For video translation, temporal consistency remains a critical metric. AECM stability in dynamic scenes is validated through inference on a self-collected visible-infrared drone video dataset. We provide comparative videos and GIF animations on our project page (https://github.com/Tracygc/AECM). These results demonstrate that AECM significantly enhances temporal coherence by reducing interframe jitter and flickering artifacts.

2) Quantitative Results: As shown in Tables I–IV, AECM achieves superior performance across all paired and unpaired benchmarks. AECM consistently yields the lowest FID and KID scores, significantly outperforming competitors. On the paired AVIID3 dataset, AECM surpasses the second-best method, DRIT, by margins of 41.46681 in FID and 0.019648 in KID. For Day-DroneVehicle and Night-DroneVehicle, AECM maintains this dominance, exceeding DR-AVIT and USTNet in both metrics. Furthermore, on the unpaired AVIID dataset, AECM outperforms CycleGAN with margins of 27.2993 for FID and 0.014521 for KID, confirming its robustness in diverse scenarios.

Analysis of perceptual realism (LPIPS), structural fidelity (SSIM), and diversity (IS) further highlights the advantages of AECM over diffusion-based models like DMT and PID. While diffusion models may yield competitive SSIM scores, they exhibit high perceptual distances (LPIPS). This discrepancy indicates that diffusion models generate high-frequency noise or artificial textures deviating from authentic thermal signatures. In contrast, AECM achieves the lowest LPIPS of 0.1494 on AVIID3paired8 while maintaining a high SSIM of 0.5619. By leveraging the multiscale PatchNCE mechanism, AECM disentangles content from style to suppress artifacts and align generated textures with authentic thermal signatures.

Furthermore, metric inflation in Inception Score (IS) is observed in methods such as MUNIT and DSMAP. As shown in Table III and Fig. 12, these methods produce IS values that exceed the real dataset (2.789533 ± 0.28418). This result indicates the generation of artificial saliency rather than

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TABLE I

QUANTITATIVE RESULTS OF DIFFERENT TRANSLATION METHODS FOR AVIT. BOLD INDICATES THE BEST RESULTS

Methods	FID ↓			KID ↓		
	AVIID3-paired8	Day-DroneVehicle	Night-DroneVehicle	AVIID3-paired8	Day-DroneVehicle	Night-DroneVehicle
ACLGAN [66]	103.11080	76.56698	127.60075	0.037753 ± 0.00280	0.035484 ± 0.00107	0.110596 ± 0.00201
CUT [23]	106.10458	57.16955	71.16025	0.024857 ± 0.00172	0.021154 ± 0.00086	0.060111 ± 0.00174
Cycle-GAN [27]	138.56059	94.32537	73.24234	0.069115 ± 0.00286	0.038125 ± 0.00375	0.058677 ± 0.00163
DRIT [22]	94.04422	47.09325	59.19516	0.020731 ± 0.00143	0.014051 ± 0.00071	0.042221 ± 0.00141
DSMAP [67]	141.97429	82.54184	89.67733	0.067294 ± 0.00346	0.052599 ± 0.00190	0.075133 ± 0.00284
DR-AVIT [1]	126.42901	31.21323	33.13756	0.053128 ± 0.00377	0.003923 ± 0.00034	0.013826 ± 0.00090
MUNIT [21]	114.10499	46.62198	51.15162	0.035898 ± 0.00347	0.015697 ± 0.00081	0.032386 ± 0.00129
USTNet [5]	105.40381	32.45447	25.25158	0.024711 ± 0.00285	0.005262 ± 0.00041	0.009781 ± 0.00006
DMT [68]	131.07320	94.02102	103.15512	0.071782 ± 0.00319	0.057772 ± 0.00109	0.088719 ± 0.00165
PID [69]	144.11531	78.04525	82.30320	0.061236 ± 0.00287	0.036711 ± 0.00096	0.062894 ± 0.00160
AECM (ours)	**52.57741**	**22.17796**	**13.25120**	**0.001083 ± 0.00085**	**0.000687 ± 0.00018**	**0.001350 ± 0.00033**

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT TRANSLATION METHODS FOR AVIT. BOLD INDICATES THE BEST RESULTS

Methods	LPIPS ↓		SSIM ↑		IS → Real	
	Day-DroneVehicle	Night-DroneVehicle	Day-DroneVehicle	Night-DroneVehicle	Day-DroneVehicle	Night-DroneVehicle
ACLGAN [66]	0.3487 ± 0.0761	0.4004 ± 0.0699	0.2453 ± 0.1289	0.2932 ± 0.0950	5.712436 ± 0.10468	5.535046 ± 0.09884
CUT [23]	0.2664 ± 0.0495	0.2980 ± 0.0612	0.3516 ± 0.1119	0.3293 ± 0.1028	5.545736 ± 0.30334	4.082665 ± 0.13607
Cycle-GAN [27]	0.2386 ± 0.0428	0.5675 ± 0.0702	0.3541 ± 0.1122	0.1141 ± 0.0448	5.193884 ± 0.35971	4.590387 ± 0.21375
DRIT [22]	0.2092 ± 0.0484	0.3114 ± 0.0470	0.4031 ± 0.1226	0.1949 ± 0.1121	5.909015 ± 0.17384	4.298703 ± 0.05525
DSMAP [67]	0.4389 ± 0.0477	0.5313 ± 0.0509	0.1709 ± 0.0790	0.2852 ± 0.0737	4.954854 ± 0.03813	3.543164 ± 0.02585
DR-AVIT [1]	0.2107 ± 0.0472	0.3243 ± 0.0812	0.3091 ± 0.0993	0.2576 ± 0.1070	5.778063 ± 0.08029	4.721201 ± 0.04170
MUNIT [21]	0.3229 ± 0.0500	0.2298 ± 0.0885	0.4113 ± 0.1080	0.4476 ± 0.0667	5.822427 ± 0.12202	4.570038 ± 0.05588
USTNet [5]	0.1983 ± 0.0507	0.2720 ± 0.0601	0.4542 ± 0.1465	0.3959 ± 0.1133	5.607312 ± 0.36865	4.041892 ± 0.16505
DMT [68]	0.4645 ± 0.0690	0.5176 ± 0.0952	0.4079 ± 0.0998	0.4181 ± 0.0795	4.841405 ± 0.25851	4.188991 ± 0.14717
PID [69]	0.3153 ± 0.0607	0.3898 ± 0.0769	0.3729 ± 0.1030	0.2475 ± 0.1030	5.622748 ± 0.21248	4.480416 ± 0.13049
AECM (ours)	**0.1689 ± 0.0461**	**0.2286 ± 0.0602**	**0.4604 ± 0.1504**	**0.4272 ± 0.1198**	**5.154103 ± 0.20726**	**4.525745 ± 0.10143**
Real IR images	0.0000 ± 0.0000	0.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	5.161948 ± 0.22675	4.533995 ± 0.11094

TABLE III

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON AVIID3-PAIRED8

Methods	LPIPS ↓	SSIM ↑	IS → Real
ACLGAN [66]	0.2177 ± 0.0597	0.5402 ± 0.1013	2.650927 ± 0.10661
CUT [23]	0.2612 ± 0.0337	0.4007 ± 0.0647	2.840571 ± 0.38762
Cycle-GAN [27]	0.3015 ± 0.0446	0.3758 ± 0.0639	3.040725 ± 0.19576
DRIT [22]	0.2544 ± 0.0329	0.4277 ± 0.0647	2.297893 ± 0.12643
DSMAP [67]	0.2566 ± 0.0386	0.5248 ± 0.0779	3.468113 ± 0.13059
DR-AVIT [1]	0.2871 ± 0.0434	0.4019 ± 0.0686	3.189783 ± 0.08072
MUNIT [21]	0.4944 ± 0.1086	0.1225 ± 0.1303	3.790277 ± 0.17566
USTNet [5]	0.2372 ± 0.0272	0.4707 ± 0.0672	2.960642 ± 0.19939
DMT [68]	0.2934 ± 0.0375	0.5601 ± 0.0585	3.034083 ± 0.34509
PID [69]	0.3780 ± 0.0358	0.3762 ± 0.0752	2.591242 ± 0.20891
AECM (ours)	**0.1494 ± 0.0316**	**0.5619 ± 0.0701**	**2.690590 ± 0.17805**
Real IR images	0.0000 ± 0.0000	1.0000 ± 0.0000	2.789533 ± 0.28418

TABLE IV

RESULTS OF DIFFERENT METHODS ON AVIID3-UNPAIRED8

Methods	FID ↓	KID ↓	IS → Real
ACLGAN [66]	109.4738	0.047430 ± 0.00366	2.717711 ± 0.14116
CUT [23]	95.9205	0.023004 ± 0.00202	2.648657 ± 0.38748
Cycle-GAN [27]	86.3111	0.016628 ± 0.00167	2.836968 ± 0.31609
DRIT [22]	105.8349	0.026952 ± 0.00262	3.185924 ± 0.22441
DSMAP [67]	133.9178	0.054089 ± 0.00530	3.878793 ± 0.13554
DR-AVIT [1]	94.3793	0.024124 ± 0.00220	3.127351 ± 0.09659
MUNIT [21]	122.6785	0.047460 ± 0.00358	3.683877 ± 0.09000
USTNet [5]	109.9623	0.034701 ± 0.00372	2.968096 ± 0.32469
DMT [68]	228.5419	0.169965 ± 0.00628	3.502510 ± 0.19279
PID [69]	144.4689	0.066823 ± 0.00287	2.583609 ± 0.17090
AECM (ours)	**59.0118**	**0.002107 ± 0.00082**	**2.728958 ± 0.24746**
Real IR images	–	–	2.773519 ± 0.32519

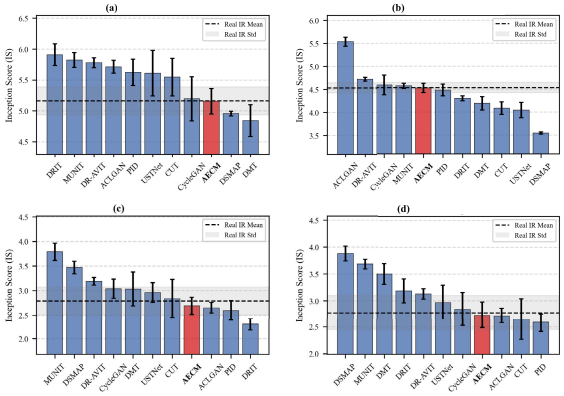


Fig. 12. Comparison of IS on (a) Day-DroneVehicle, (b) Night-DroneVehicle, (c) AVIID3-paired8, and (d) AVIID3-unpaired8. The dashed line and gray area represent the mean and standard deviation of real infrared images.

authentic data distributions. Conversely, AECM achieves an IS of 2.690590, falling within the GT confidence interval. This alignment, combined with superior LPIPS results, confirms that AECM generates diverse samples without mode collapse or artifact-induced inflation. These results validate that multiscale CL enables AECM to preserve global geometric structures and synthesize fine-grained infrared textures consistent with human perception.

Table V reports the inference speed comparisons. AECM demonstrates a significant latency advantage over diffusion-based models like PID and DMT, which require iterative denoising passes. As a one-step feed-forward generator, AECM is highly suitable for deployment on resource-

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TABLE V
COMPARISON OF EFFICIENCY OF DIFFERENT METHODS

Methods	ACLGAN	CUT	Cycle-GAN	DRIT	DSMAP
Time (ms) ↓	62.0	45.2	46.2	65.0	70.0
DR-AVIT	MUNIT	USTNet	DMT	PID	**AECM**
55.0	68.5	60.0	117.1	76.1	45.0

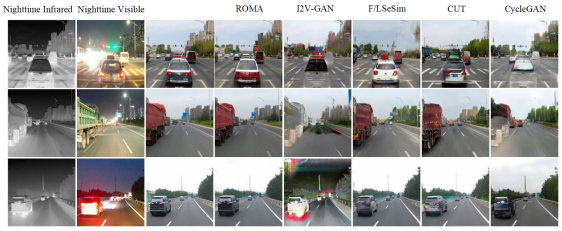


Fig. 13. Results comparison with different methods on ICL.

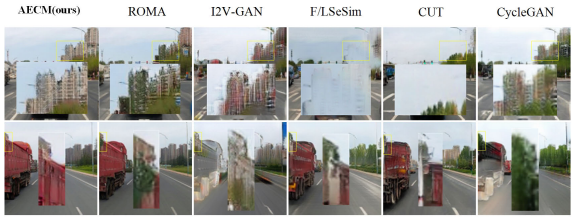


Fig. 14. Comparison of details in generated images from different algorithms.

constrained UAV edge devices. The multiscale ViT discriminator and auxiliary Swin Transformer blocks are utilized only during training and removed during inference to ensure efficiency. Furthermore, the shifted-window mechanism in the swin transformer blocks reduces computational complexity to a linear scale, enabling rapid translation without compromising quality.

D. Comparison Experiments for IVIVT

To evaluate the proposed AECM for IVIVT, extensive comparative experiments are conducted against SOTA methods.

1) Qualitative Results: Fig. 13 presents qualitative translation results on ICL. AECM demonstrates superior preservation of structural integrity and color fidelity compared to competing frameworks. While CycleGAN and I2V-GAN produce nearly transparent vehicles, CUT and F/LSeSim fail to reconstruct peripheral building structures. Rome exhibits semantic errors by misclassifying distant buildings as road signs or trees. In contrast, AECM maintains accurate foreground vehicle structures and realistic background details. Fig. 14 provides enlarged views to further highlight these differences. In the second row, whereas other methods introduce leaf-like texture artifacts or structural distortions into truck boundaries, AECM generates precise outlines. Although all algorithms identify distant small vehicles in the third row, only AECM and Rome avoid the geometric deformations observed in CycleGAN, CUT, and F/LSeSim. Temporal consistency is evaluated using consecutive frames from the City subset as shown in Fig. 15. AECM, Rome, and I2V-GAN maintain stable structures across sequences, though I2V-GAN deviates from natural daytime



Fig. 15. Translated consecutive frames of different methods on highway.

TABLE VI
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON ICL. BOLD INDICATES THE BEST RESULTS

Methods	FID ↓		KID ↓	
	City	Highway	City	Highway
Cycle-GAN [27]	74.50	53.02	0.03178	0.02291
CUT [23]	56.64	37.11	0.01587	0.00834
F-LSeSim [35]	54.24	41.60	0.01971	0.01441
I2V-GAN [51]	99.92	239.24	0.06018	0.26524
ROMA [52]	52.68	33.70	0.01243	0.00704
AECM (ours)	**51.83**	**32.67**	**0.01215**	**0.00679**

TABLE VII
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON IRVI. BOLD INDICATES THE BEST RESULTS

Methods	FID ↓		KID ↓	
	Traffic	rooftop_01	Traffic	rooftop_01
Cycle-GAN [27]	65.43	186.10	0.05026	0.11507
CUT [23]	42.60	151.38	0.02563	0.09308
F-LSeSim [35]	44.27	153.10	0.03026	**0.08756**
I2V-GAN [51]	79.22	251.24	0.06734	0.22356
ROMA [52]	41.96	144.05	**0.02444**	0.09666
AECM (ours)	**41.69**	**142.29**	0.02499	0.08881

characteristics. Other methods exhibit interframe deformations that cause flickering in synthesized videos. By incorporating cross-domain similarity temporal loss, AECM effectively ensures temporal coherence and minimizes jitter or deformation issues in dynamic scenes.

2) Quantitative Results: Tables VI and VII present FID and KID scores for ICL and IRVI benchmarks. On ICL subdatasets, AECM outperforms competing methods by an average of 25.43% in FID and 44.85% in KID. On IRVI subdatasets, AECM achieves improvements of 17.34% in FID and 22.52% in KID. Except for the Traffic subset in IRVI,

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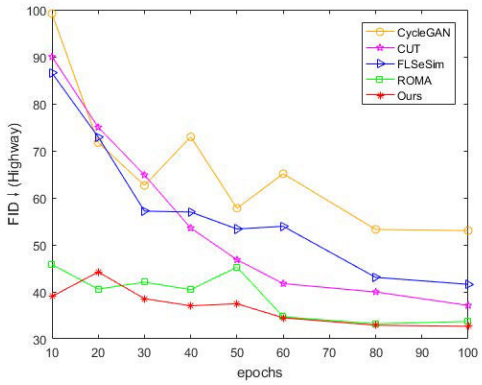


Fig. 16. Line chart of FID for different training epochs.

TABLE VIII
TRAINING TIME OF DIFFERENT TRANSLATION METHODS ON HIGHWAY

Methods		time/h↓ 100 epochs (Highway)
video	I2V-GAN	49.42
	ROMA	39.27
	AECM (ours)	**36.42**

the results of AECM surpass those of Rome. Moreover, our method achieves the best FID results on all four subdatasets. These results demonstrate the superior performance of AECM in infrared-to-visible video translation.

To investigate the temporal efficiency of our method, we evaluate FID at various training epochs on the Highway subset of ICL. As shown in Fig. 16, by the tenth epoch, AECM reduces its FID to 39.03 compared to 45.80 for Rome. By the 100th epoch, AECM achieves the optimal FID and demonstrates faster stability than competing frameworks. We also compare the total training time of the translation algorithms for 100 epochs. As shown in Table VIII, our AECM is the most time-efficient in video translation, being 26.53% faster than I2V-GAN and 7.26% faster than Rome.

The superior performance of our AECM is attributed to the following factors. First, the attention-enhanced ViT discriminator, through adversarial learning with the generator, improves the detail quality and realism of the generated images. Second, the shifted-attention cross-similarity loss enhances the ability to maintain the global structure of images by introducing the shifted-attention computation of the swin transformer. Finally, the learnable-weight-optimized PatchNCE loss further refines color and texture accuracy while mitigating local distortion issues caused by CL. These components enable AECM to outperform other methods in structural preservation and color restoration.

E. Ablation Study

To comprehensively evaluate the contribution of each component in the proposed AECM, we conduct systematic ablation studies on AVIT and IVIVT.

1) Effects of Each Part of AECM on AVIID for AVIT: To evaluate the contribution of each component in AVIT, we conduct ablation studies by incrementally removing key modules, including the attention-enhanced ViT discriminator, the shifted-window cross-similarity loss $\mathcal{L}_{swin_sim}$, the learnable-weight-optimized PatchNCE loss $\mathcal{L}_{LWO_NCE}$, the temporal consistency loss $\mathcal{L}_{temporal}$, and the generator. Table IX presents the quantitative results on the AVIID dataset, covering both the unpaired and paired data subsets.

TABLE IX
ABLATION STUDY ON THE EFFECTIVENESS OF COMPONENTS ON AVIID

Methods	FID ↓ unpaired	FID ↓ paired	KID ↓ unpaired	KID ↓ paired
Ours (w/o ViT)	347.02	341.09	0.360063	0.364344
Ours (w/o D_a)	60.066	55.686	0.001468	0.001263
Ours (w/o D_m)	64.563	60.130	0.003858	0.003258
Ours (w/o $\mathcal{L}_{swin_sim}$)	62.017	54.708	0.002205	0.001712
Ours (w/o $\mathcal{L}_{LWO_NCE}$)	65.258	57.686	0.003260	0.001711
Ours (w/o $\mathcal{L}_{temporal}$)	61.749	56.503	0.001799	0.000517
Ours (G (Unet256))	64.080	57.077	0.004669	0.003355
Ours (all)	**59.012**	**52.577**	**0.002107**	**0.001083**

When the attention-enhanced ViT discriminator is replaced with a conventional CNN-based discriminator (denoted as w/o ViT), FID and KID scores increase, indicating performance degradation. This result confirms that the ViT-based discriminator captures long-range dependencies and global contextual information, providing more accurate adversarial feedback. Removing the attention mechanism within the discriminator (w/o D_a) also leads to performance deterioration, highlighting the importance of the integrated attention module for enabling the discriminator to focus on semantically critical regions and enhance image quality. Removing the multiscale ViT discriminator (w/o D_m) results in further degradation, suggesting that adversarial loss alone is insufficient to enforce spatial consistency constraints. Removing temporal consistency loss $\mathcal{L}_{temporal}$ also leads to an increase in FID on both paired and unpaired subsets. Although the target task is static image translation, the dataset consists of frames extracted from continuous aerial videos, which often contain motion artifacts or transient sensor noise. Enforcing consistency between adjacent frames during training enables $\mathcal{L}_{temporal}$ to act as a regularizer that suppresses transient noise and guides the generator to learn stable intrinsic structural representations.

Additionally, maximizing mutual information through CL effectively preserves structural integrity while permitting stylistic variation. Specifically, removing the shifted-window cross-similarity loss (w/o $\mathcal{L}_{swin_sim}$) reduces the model's ability to capture content and structural correspondences between source and target domains. This diminishes multilevel feature alignment from global to local scales and reduces cross-domain consistency, impairing generation quality. Similarly, the inferior results of the variant without $\mathcal{L}_{LWO_NCE}$ demonstrate that the weight adjustment strategy in PatchNCE is crucial for maintaining feature-level correspondence and textural detail. Regarding the generator, the ResNet generator outperforms the U-Net variant by better capturing the complex mappings required for translation.

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TABLE X
EFFECTS OF THREE KEY COMPONENTS OF AECM FOR IVIVT

Methods	FID ↓		KID ↓	
	ICL Highway	IRVI rooftop_01	ICL Highway	IRVI rooftop_01
only D_a	235.54	260.28	0.29955	0.27451
only $\mathcal{L}_{swin_sim}$	39.35	147.34	0.00917	0.09405
only $\mathcal{L}_{LWO_NCE}$	37.28	143.28	0.00882	0.08991
Ours (all)	**32.67**	**142.29**	**0.00679**	**0.08881**

The performance of AECM originates from the global modeling capacity of the vision transformer (ViT). Unlike CNNs restricted by local receptive fields, the self-attention mechanism in ViTs aggregates information across all patches to capture structural and semantic interdependencies essential for complex aerial imagery. This global perspective enhances two critical components. First, the ViT-based discriminator evaluates local texture fidelity alongside global layout coherence to provide a more rigorous adversarial signal. Second, ViT-extracted features establish robust semantic correspondences for CL based on high-order global context rather than local similarities. Consequently, this ensures superior content preservation and realistic texture synthesis during cross-modality translation.

Table IX confirms that while the ViT-based method establishes a high-performance benchmark, individual algorithmic components are essential for optimal results. Replacing the ViT discriminator causes the FID to surge to 347.02, verifying the necessity of a global receptive field for cross-modality translation. LWO-NCE loss emerges as the most significant algorithmic contributor to generative precision in AECM. Omitting $\mathcal{L}_{LWO_NCE}$ results in an FID increase of 6.246 on the unpaired AVIID subset, representing a larger degradation than the 1.054 increase observed when removing the attention mechanism D_a. This evidence suggests that adaptive feature-level correspondence is more critical for high-fidelity synthesis than the marginal gains from the discriminator attention module. Furthermore, $\mathcal{L}_{swin_sim}$ and $\mathcal{L}_{temporal}$ provide essential structural anchoring to ensure stability and consistency across frames. These findings verify that the synergy between the ViT-based adversarial signal and the optimized CL objective enables AECM to maintain superior structural integrity and realistic infrared textures.

2) Effects of Three Key Components of AECM for IVIVT: To evaluate the effectiveness of AECM designs for IVIVT, we perform ablation experiments examining three key components. The ablation results in Table X further highlight the critical role of LWO-NCE loss. Although the attention-enhanced discriminator D_a provides a rigorous adversarial signal, employing D_a in isolation leads to excessive FID and KID scores due to the absence of explicit content constraints. This structural collapse confirms that adversarial learning alone is insufficient for maintaining semantic integrity. The integration of $\mathcal{L}_{LWO_NCE}$ facilitates the most significant performance gain among all individual components. Specifically, on the ICL Highway subset, $\mathcal{L}_{LWO_NCE}$ achieves an FID of 37.28, which substantially outperforms the only-D_a and only-$\mathcal{L}_{swin_sim}$ configurations. These findings demonstrate that adaptive and learnable-weight CL is the primary driver for high-fidelity texture synthesis. Furthermore, the optimal results of AECM verify the synergistic benefits of combining global structural anchoring via $\mathcal{L}_{swin_sim}$ with the precise feature-level correspondence enforced by $\mathcal{L}_{LWO_NCE}$.

3) Analysis of Token Embedding Selection: To validate the effectiveness of specific token embeddings, the performance of various encoding block combinations is evaluated on AVIID3. As summarized in Tables XI and XII, and Fig. 17, the ViT encodes representations ranging from local details to global semantics. The shallow first block preserves high-frequency textures essential for small object geometry, whereas the

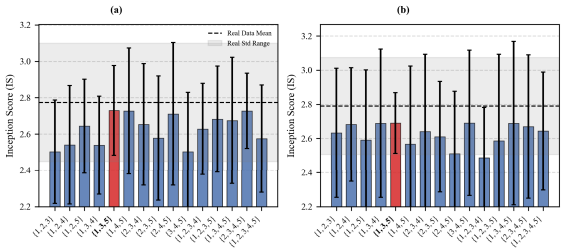


Fig. 17. Comparison of IS on different encoding blocks. The red bars represent the selected configuration, which demonstrates the closest statistical alignment with the distribution of real infrared images (dashed line). (a) AVIID3-unpaired8. (b) AVIID3-paired8.

TABLE XI
DIFFERENT ENCODING BLOCKS ON AVIID3-UNPAIRED8

Encoding blocks	FID ↓	KID ↓	IS → Real
{1st, 2nd, and 3rd}	68.374	0.00425 ± 0.0010	2.5022 ± 0.285
{1st, 2nd, and 4th}	66.808	0.00444 ± 0.0009	2.5402 ± 0.326
{1st, 2nd, and 5th}	62.071	0.00286 ± 0.0007	2.6433 ± 0.258
{1st, 3rd, and 4th}	66.800	0.00402 ± 0.0010	2.5383 ± 0.270
{1st, 3rd, and 5th}	**59.012**	**0.00211 ± 0.0008**	**2.7290 ± 0.247**
{1st, 4th, and 5th}	62.242	0.00296 ± 0.0007	2.7267 ± 0.347
{2nd, 3rd, and 4th}	65.489	0.00377 ± 0.0008	2.6529 ± 0.334
{2nd, 3rd, and 5th}	61.744	0.00249 ± 0.0006	2.5778 ± 0.342
{2nd, 4th, and 5th}	61.174	0.00116± 0.0007	2.6130 ± 0.338
{3rd, 4th, and 5th}	60.897	0.00219 ± 0.0007	2.5025 ± 0.327
{1st, 2nd, 3rd, and 4th}	68.735	0.00343 ± 0.0010	2.6277 ± 0.251
{1st, 2nd, 3rd, and 5th}	63.554	0.00293 ± 0.0007	2.6822 ± 0.291
{1st, 3rd, 4th, and 5th}	66.011	0.00253 ± 0.0008	2.6741 ± 0.347
{2nd, 3rd, 4th, and 5th}	62.791	0.00262 ± 0.0009	2.7271 ± 0.207
{1st, 2nd, 3rd, 4th, and 5th}	63.508	0.00268 ± 0.0008	2.5750 ± 0.295

TABLE XII
DIFFERENT ENCODING BLOCKS ON AVIID3-PAIRED8

Encoding blocks	FID ↓	LPIPS ↓	SSIM ↑
{1st, 2nd, and 3rd}	66.857	0.2036 ± 0.0411	0.4518 ± 0.0753
{1st, 2nd, and 4th}	62.397	0.1929 ± 0.0396	0.4617 ± 0.0730
{1st, 2nd, and 5th}	56.576	0.1816 ± 0.0373	0.4874 ± 0.0706
{1st, 3rd, and 4th}	61.067	0.1806 ± 0.0359	0.4874 ± 0.0719
{1st, 3rd, and 5th}	**52.577**	**0.1494 ± 0.0316**	**0.5319 ± 0.0701**
{1st, 4th, and 5th}	55.668	0.1858 ± 0.0372	0.4665 ± 0.0708
{2nd, 3rd, and 4th}	61.139	0.1940 ± 0.0389	0.4811 ± 0.0711
{2nd, 3rd, and 5th}	57.584	0.1796 ± 0.0342	0.4516 ± 0.1062
{2nd, 4th, and 5th}	57.590	0.1903 ± 0.0402	0.4865 ± 0.0707
{3rd, 4th, and 5th}	57.729	0.1819 ± 0.0360	0.4889 ± 0.0688
{1st, 2nd, 3rd, and 4th}	62.093	0.1939 ± 0.0409	0.4654 ± 0.0690
{1st, 2nd, 3rd, and 5th}	57.443	0.1832 ± 0.0362	0.4880 ± 0.0681
{1st, 3rd, 4th, and 5th}	56.909	0.1797 ± 0.0353	0.4903 ± 0.0695
{2nd, 3rd, 4th, and 5th}	58.069	0.1816 ± 0.0369	0.4889 ± 0.0703
{1st, 2nd, 3rd, 4th, and 5th}	56.765	0.1827 ± 0.0364	0.4882 ± 0.0686

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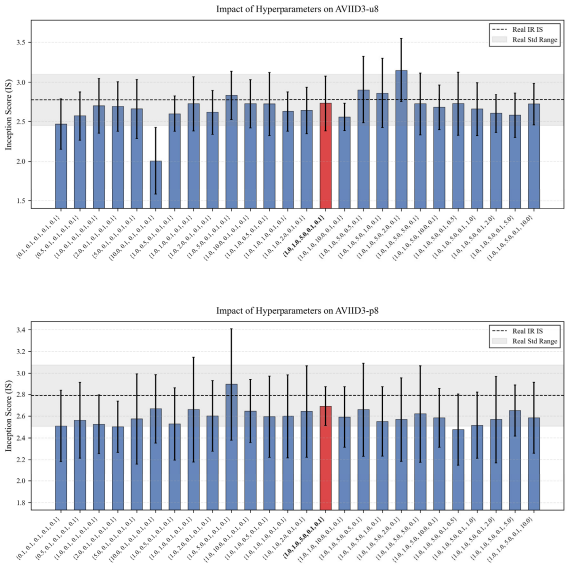


Fig. 18. Comparison of IS across different hyperparameter configurations. The red bars represent the selected configuration, which demonstrates the closest statistical alignment with the distribution of real infrared images (dashed line).

middle third block constructs part-level features. The deeper 5*th* block captures global context and semantic relationships. Selecting the fifth block instead of the deepest layer prevents excessive abstraction and preserves the spatial precision required for dense prediction. Quantitative results corroborate this theoretical analysis. The first, third, and fifth configuration achieves an optimal balance between perceptual quality and structural fidelity. On the unpaired dataset in Table XI, this widely spaced configuration reaches the lowest FID of 59.012 and KID of 0.00211, surpassing adjacent-layer combinations like first, second, and third that exhibit redundant features. On the paired dataset in Table XII, this selection yields the highest SSIM of 0.5319 and the lowest LPIPS of 0.1494. These findings confirm that multiscale fusion facilitates superior structural consistency compared to deeper combinations like third, fourth, and fifth. Furthermore, the IS for the first, third, and fifth setup is statistically closest to the real infrared benchmark in Fig. 17 and Table XI. This alignment verifies that the selected multiscale strategy synthesizes the most realistic thermal textures while maintaining structural integrity.

4) Sensitivity Analysis of Hyperparameters: To identify the optimal configuration for loss coefficients $\lambda_1, \lambda_2, \lambda_3, \lambda_4, and \lambda_5$ in (9) and (10), sensitivity analysis is conducted on AVIID3 benchmarks. Results in Tables XIII and XIV demonstrate tradeoffs among structural fidelity, adversarial realism, and regularization. Empirical evidence suggests that $\{1.0, 1.0, 5.0, 0.1, 0.1\}$ is the optimal coefficient set. Structural consistency is the dominant constraint as indicated by $\lambda_3 = 5.0$. Increasing λ_3 from 1.0 to 5.0 improves SSIM from 0.4915 to 0.5319 and yields the lowest FID of 52.577. Preserving the geometric integrity of small objects is critical for UAV-view V2I translation. A higher λ_3 value stabilizes the model and prevents structural distortions during domain transfer. Adversarial components with $\lambda_1 = 1.0$ and $\lambda_2 = 1.0$ regulate the multiscale and attention-based discriminators to guide the synthesis of realistic thermal signatures. Analysis shows that lower values lead to blurred outputs and suboptimal alignment, while excessively high values cause training instability. Setting both coefficients to 1.0 ensures a balanced adversarial signal for capturing fine-grained thermal distributions. Regarding regularization, the learnable-weight-optimized (LWO-NCE) loss ($\lambda_4 = 0.1$) and temporal loss

TABLE XIII
DIFFERENT CONFIGURATIONS OF HYPERPARAMETERS ON AVIID3-U8

$\{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\}$	FID ↓	KID ↓	IS → Real
{0.1, 0.1, 0.1, 0.1, 0.1}	84.444	0.00907 ± 0.0011	2.4681 ± 0.316
{0.5, 0.1, 0.1, 0.1, 0.1}	64.286	0.00244 ± 0.0008	2.5717 ± 0.305
{1.0, 0.1, 0.1, 0.1, 0.1}	61.001	0.00222 ± 0.0009	2.6990 ± 0.347
{2.0, 0.1, 0.1, 0.1, 0.1}	62.036	0.00230 ± 0.0008	2.6918 ± 0.314
{5.0, 0.1, 0.1, 0.1, 0.1}	61.781	0.00292 ± 0.0010	2.6607 ± 0.373
{10.0, 0.1, 0.1, 0.1, 0.1}	108.032	0.02616 ± 0.0021	2.0035 ± 0.421
{1.0, 0.5, 0.1, 0.1, 0.1}	67.872	0.00346 ± 0.0009	2.5979 ± 0.222
{1.0, 1.0, 0.1, 0.1, 0.1}	61.329	0.00301 ± 0.0010	2.7245 ± 0.343
{1.0, 2.0, 0.1, 0.1, 0.1}	67.741	0.00397 ± 0.0011	2.6181 ± 0.279
{1.0, 5.0, 0.1, 0.1, 0.1}	66.512	0.00322 ± 0.0009	2.8296 ± 0.306
{1.0, 10.0, 0.1, 0.1, 0.1}	61.355	0.00258 ± 0.0011	2.7262 ± 0.306
{1.0, 1.0, 0.5, 0.1, 0.1}	61.414	0.00229 ± 0.0009	2.7232 ± 0.399
{1.0, 1.0, 1.0, 0.1, 0.1}	67.723	0.00347 ± 0.0009	2.6278 ± 0.250
{1.0, 1.0, 2.0, 0.1, 0.1}	61.430	0.00278 ± 0.0007	2.6410 ± 0.296
{1.0, 1.0, 5.0, 0.1, 0.1}	**59.012**	**0.00211 ± 0.0008**	**2.7290 ± 0.347**
{1.0, 1.0, 10.0, 0.1, 0.1}	64.099	0.00245 ± 0.0008	2.5567 ± 0.172
{1.0, 1.0, 5.0, 0.5, 0.1}	64.141	0.00301 ± 0.0009	2.9036 ± 0.421
{1.0, 1.0, 5.0, 1.0, 0.1}	78.209	0.00579 ± 0.0010	2.8619 ± 0.439
{1.0, 1.0, 5.0, 2.0, 0.1}	139.494	0.0444 ± 0.0030	3.1483 ± 0.399
{1.0, 1.0, 5.0, 5.0, 0.1}	78.850	0.00608 ± 0.0010	2.7233 ± 0.392
{1.0, 1.0, 5.0, 10.0, 0.1}	344.018	0.36833 ± 0.0074	2.6796 ± 0.284
{1.0, 1.0, 5.0, 0.1, 0.5}	64.039	0.00244 ± 0.0008	2.7248 ± 0.399
{1.0, 1.0, 5.0, 0.1, 1.0}	67.691	0.00311 ± 0.0009	2.6575 ± 0.336
{1.0, 1.0, 5.0, 0.1, 2.0}	65.173	0.00281 ± 0.0008	2.6041 ± 0.241
{1.0, 1.0, 5.0, 0.1, 5.0}	71.897	0.00556 ± 0.0011	2.5805 ± 0.282
{1.0, 1.0, 5.0, 0.1, 10.0}	76.958	0.00724 ± 0.0012	2.7207 ± 0.265

TABLE XIV
DIFFERENT CONFIGURATIONS OF HYPERPARAMETERS ON AVIID3-P8

$\{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\}$	FID ↓	LPIPS ↓	SSIM ↑
{0.1, 0.1, 0.1, 0.1, 0.1}	62.605	0.1970 ± 0.0356	0.4737 ± 0.0743
{0.5, 0.1, 0.1, 0.1, 0.1}	61.017	0.1923 ± 0.0338	0.4752 ± 0.0730
{1.0, 0.1, 0.1, 0.1, 0.1}	61.958	0.1983 ± 0.0358	0.4505 ± 0.0746
{2.0, 0.1, 0.1, 0.1, 0.1}	63.063	0.1942 ± 0.0356	0.4565 ± 0.0710
{5.0, 0.1, 0.1, 0.1, 0.1}	64.389	0.1988 ± 0.0363	0.4675 ± 0.0760
{10.0, 0.1, 0.1, 0.1, 0.1}	65.777	0.2005 ± 0.0367	0.4693 ± 0.0747
{1.0, 0.5, 0.1, 0.1, 0.1}	57.325	0.1646 ± 0.0312	0.5121 ± 0.0705
{1.0, 1.0, 0.1, 0.1, 0.1}	55.084	0.1607 ± 0.0312	0.5138 ± 0.0719
{1.0, 2.0, 0.1, 0.1, 0.1}	54.311	0.1562 ± 0.0310	0.5210 ± 0.0714
{1.0, 5.0, 0.1, 0.1, 0.1}	54.146	0.1525 ± 0.0303	0.5210 ± 0.0725
{1.0, 10.0, 0.1, 0.1, 0.1}	53.536	0.1550 ± 0.0285	0.5201 ± 0.0736
{1.0, 1.0, 0.5, 0.1, 0.1}	56.090	0.1653 ± 0.0306	0.5166 ± 0.0701
{1.0, 1.0, 1.0, 0.1, 0.1}	54.470	0.1655 ± 0.0326	0.4915 ± 0.0706
{1.0, 1.0, 2.0, 0.1, 0.1}	54.879	0.1707 ± 0.0345	0.5026 ± 0.0700
{1.0, 1.0, 5.0, 0.1, 0.1}	**52.577**	**0.1494 ± 0.0316**	**0.5319 ± 0.0701**
{1.0, 1.0, 10.0, 0.1, 0.1}	61.652	0.1929 ± 0.0385	0.4755 ± 0.0704
{1.0, 1.0, 5.0, 0.5, 0.1}	59.489	0.1908 ± 0.0373	0.4585 ± 0.0737
{1.0, 1.0, 5.0, 1.0, 0.1}	62.019	0.1977 ± 0.0376	0.4510 ± 0.0722
{1.0, 1.0, 5.0, 2.0, 0.1}	67.586	0.2115 ± 0.0375	0.4496 ± 0.0756
{1.0, 1.0, 5.0, 5.0, 0.1}	78.862	0.2398 ± 0.0429	0.4173 ± 0.0779
{1.0, 1.0, 5.0, 10.0, 0.1}	364.669	0.5839 ± 0.0463	0.2441 ± 0.0720
{1.0, 1.0, 5.0, 0.1, 0.5}	56.641	0.1825 ± 0.0360	0.4906 ± 0.0680
{1.0, 1.0, 5.0, 0.1, 1.0}	58.188	0.1843 ± 0.0360	0.4881 ± 0.0694
{1.0, 1.0, 5.0, 0.1, 2.0}	59.969	0.1917 ± 0.0352	0.4809 ± 0.0717
{1.0, 1.0, 5.0, 0.1, 5.0}	69.195	0.2170 ± 0.0359	0.4498 ± 0.0727
{1.0, 1.0, 5.0, 0.1, 10.0}	73.845	0.2352 ± 0.0349	0.4275 ± 0.0687

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TABLE XV
MAP@50 FOR OBJECT DETECTION IN INFRARED IMAGES

Methods	mAP@50 (%) ↑	
	Day-DroneVehicle	Night-DroneVehicle
UNIT [70]	71.7	57.0
CUT [23]	82.3	63.9
Cycle-GAN [27]	83.1	68.5
DRIT [22]	83.1	65.8
MUNIT [21]	82.8	57.3
DCLGAN [71]	79.5	63.5
MSFD [6]	83.6	72.0
DR-AVIT [1]	83.2	69.7
ATEN (Ours)	**88.2**	**85.7**



Fig. 19. Results of object detection using YOLOv5.

($\lambda_5 = 0.1$) serve as auxiliary refinement terms. Increasing these beyond 1.0 degrades perceptual quality due to excessive regularization conflicting with adversarial objectives. Distributional alignment with real data is further validated in Fig. 18. The selected configuration achieves Inception Scores (IS) of 2.7290 ± 0.347 and 2.6906 ± 0.178 for the unpaired and paired subsets, respectively. These values are statistically closest to the real infrared benchmarks of 2.7735 and 2.7895. These results confirm the effectiveness of the $\{1.0, 1.0, 5.0, 0.1, 0.1\}$ configuration across all translation experiments.

F. Downstream Applications

To further validate the downstream applicability of our AECM, we perform experiments in object detection and image fusion, demonstrating its practical effectiveness.

1) Object Detection on AVIT: Based on the results of the object detection task, as shown in Table XV, our AECM

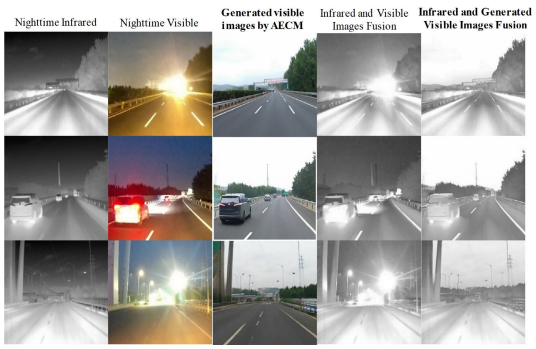


Fig. 20. Fusion results of infrared images with the other two (fusing with nighttime visible images and translated visible images).

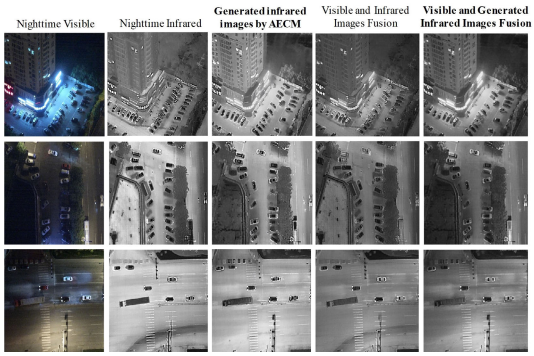


Fig. 21. Fusion results of visible images with the other two (fusing with nighttime infrared images and translated infrared images).

achieves the best mAP@50 at 88.2% and 85.7% in the generation of infrared drone images on Day-DroneVehicle and Night-DroneVehicle, respectively. Despite the challenges of Night-DroneVehicle caused by low illumination and sensor noise, AECM maintains stable performance, demonstrating its strong robustness. This advantage stems from the integrated attention mechanism within the generative adversarial network, enabling effective processing of images with multiscale objects, densely packed small targets, and complex backgrounds. Furthermore, the interaction between the offset attention cross-similarity loss and the weight-optimized PatchNCE loss balances global and local self-attention, further enhancing the quality of generated images.

2) Object Detection on IVIVT: Infrared images pose challenges due to low resolution and limited texture, while nighttime visible images are constrained by poor illumination, resulting in suboptimal detection outcomes. Therefore, object detection with translated visible images is valuable. We employ a pretrained YOLOv5 to detect vehicles in infrared, nighttime visible, and translated visible frames. Fig. 19 displays representative detection results. Both infrared and nighttime visible frames exhibit missed targets, with the latter more affected by lighting interference. In contrast, AECM-generated frames show better detection performance.

3) Image Fusion on AVIT and IVIVT: Infrared and visible image fusion enhances semantic content. To validate the versatility of AECM, bidirectional image translation and fusion are performed without prior registration. As shown

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in Figs. 20 and 21, fusing a real image with its AECM-generated counterpart yields superior results compared to using unregistered real image pairs. While intense lighting in real nighttime visible images often obscures information during fusion, AECM-generated images effectively mitigate these artifacts. This approach provides richer detail and bypasses perspective mismatches by maintaining strict structural consistency, confirming the potential of AECM for high-quality, streamlined image fusion.

V. Discussion

AECM demonstrates a significant advantage in balancing inference speed and translation quality. Although removing the multiscale ViT discriminator and auxiliary swin transformer blocks during inference facilitates deployment on UAV platforms, the training complexity and parameter count remain substantial. Further structural compression and model quantization could enhance applicability on edge devices with restricted memory. While the framework demonstrates robustness in standard aerial scenarios, its stability under extreme environmental occlusions or high sensor noise necessitates additional investigation. Future research will focus on developing lightweight transformer architectures to balance inference speed and generative quality. Furthermore, exploring unsupervised or semi-supervised domain adaptation (UDA/SSDA) techniques represents a critical direction for improving model generalization across diverse meteorological conditions.

VI. Conclusion

In this article, we propose AECM, an innovative framework for cross-modality translation of unpaired images and videos, specifically designed for aerial scenarios. By systematically integrating ViT modules, the framework effectively addresses the critical challenge of preserving content consistency during complex style transfer. The main contributions of this work are summarized as follows. First, we introduce an attention-enhanced multiscale ViT discriminator, which models global context and long-range dependencies more effectively than traditional CNN discriminators, thereby generating images with higher visual quality and more coherent structure. Second, we propose a shifted window cross-similarity loss, which enhances the model's ability to capture and transfer cross-domain semantic content by enforcing feature alignment across different scales. Third, we design a learnable-weight-optimized PatchNCE CL loss. This strategy dynamically adjusts the importance of feature blocks to optimize the PatchNCE loss, thereby preserving the detailed correspondence between input and output. These components collectively enhance AECM's capacity for content retention and style transfer. The dual-discriminator setup ensures multiscale perceptual fidelity and semantic focus, while the sliding self-attention and adaptive contrastive losses anchor structural consistency from global to local levels. This joint optimization effectively balances the tradeoff between thermophysical plausibility and realistic texture synthesis in complex aerial scenarios. Experimental results across multiple benchmarks confirm that AECM achieves lower FID and KID scores compared to existing methods, establishing its effectiveness and generalizability in complex aerial image translation tasks.

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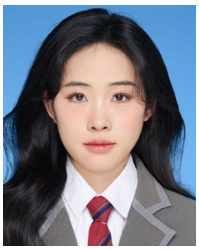
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